Logic, Language, and Philosophy

Proposition logic: validity

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Today

- Introduction
- Of properties and relations
  - Equivalence
  - Tautology, contradiction, contingency
  - Connections
- Validity
- Functional completeness
- Homework

Introduction

Recall

Truth tables can be used to:

- define connectives
- investigate systematic connections between formulas

Key features:

- referential: only truth values matter
- finite: formulas are finite objects
- mechanical: for an arbitrary distribution of truth values over the proposition letters in a formula the truth value of the formula as a whole can be uniquely determined by applying the operations specified by the connectives that occur in it

Valuations

Another way to do the same: valuations

Definition (Semantics)

A valuation V is a function from FORM_L to $\{0, 1\}$ that meets the following requirements:

1. $V(p_i) \in \{0, 1\}$, for all $p_i \in \text{VOC}_L$
2. $V(\neg\phi) = 1$ iff $V(\phi) = 0$
3. $V(\phi \wedge \psi) = 1$ iff $V(\phi) = 1$ and $V(\psi) = 1$
4. $V(\phi \vee \psi) = 1$ iff $V(\phi) = 1$ or $V(\psi) = 1$
5. $V(\phi \to \psi) = 1$ iff $V(\phi) = 0$ or $V(\psi) = 1$
6. $V(\phi \leftrightarrow \psi) = 1$ iff $V(\phi) = V(\psi)$

Putting both to use

- Study properties of, and relations between, formulas
- Study validity of argument schemes

Of valuations and truth tables

Note:

- Each row in a truth table defines a set of valuations, viz., all those that assign the same truth value to the proposition letters in the truth table
- Conversely, each valuation defines a row in a truth table when limited to the proposition letters in the truth table

So we can switch between valuations and truth tables because:

- A formula contains only a finite set of proposition letters and finite set of connectives
- The truth value of a formula depends only on the truth values of its proposition letters and on its connectives

Of properties and relations

Of properties and relations

Equivalence

Definition

Definition (Logical equivalence)

ϕ and ψ are logically equivalent iff for all valuations V: $V(\phi) = V(\psi)$

Note:

- 'Logical' equivalence to distinguish this from 'material' equivalence ($\leftrightarrow$)
- When the meaning is clear from the context we drop 'logical' / 'material'

Notation:

- 'ϕ and ψ are equivalent' is written as '$\phi \Leftrightarrow \psi$'

Checking equivalence: truth tables

$p \to q \Leftrightarrow \neg q \to \neg p$

p	q	$\neg p$	$\neg q$	$p \to q$	$\neg q \to \neg p$
1	1	0	0	1	1
1	0	0	1	0	0
0	1	1	0	1	1
0	0	1	1	1	1

$\phi \to \psi \Leftrightarrow \neg\psi \to \neg\phi$

ϕ	ψ	$\neg\phi$	$\neg\psi$	$\phi \to \psi$	$\neg\psi \to \neg\phi$
1	1	0	0	1	1
1	0	0	1	0	0
0	1	1	0	1	1
0	0	1	1	1	1

Checking equivalence: valuations

Fact (Law of Contraposition)

$\phi \to \psi \Leftrightarrow \neg\psi \to \neg\phi$

Proof.

$\Rightarrow$: Suppose $V(\phi \to \psi) = 1$ for some arbitrary V. Then we know that if $V(\phi) = 1$ then also $V(\psi) = 1$. From that it follows that $V(\neg\phi) = V(\neg\psi) = 0$ and that means that $V(\neg\psi \to \neg\phi) = 1$. Since V was chosen arbitrarily, this holds for all V

$\Leftarrow$: [by similar reasoning]

□

Useful equivalences

a. $\phi \Leftrightarrow \neg\neg\phi \Leftrightarrow \phi \wedge \phi \Leftrightarrow \phi \vee \phi \Leftrightarrow \phi \wedge (\phi \vee \psi) \Leftrightarrow \phi \vee (\phi \wedge \psi)$

b. $\neg\phi \Leftrightarrow \phi \rightarrow (\psi \wedge \neg\psi) \Leftrightarrow \phi \rightarrow \bot$

c. $\neg(\phi \vee \psi) \Leftrightarrow \neg\phi \wedge \neg\psi$ (De Morgan's Law)

d. $\neg(\phi \wedge \psi) \Leftrightarrow \neg\phi \vee \neg\psi$ (De Morgan's Law)

e. $\phi \vee \psi \Leftrightarrow \psi \vee \phi$ (commutativity of $\vee$)

f. $\phi \wedge \psi \Leftrightarrow \psi \wedge \phi$ (commutativity of $\wedge$)

g. $\phi \wedge (\psi \vee \chi) \Leftrightarrow (\phi \wedge \psi) \vee (\phi \wedge \chi)$ (distributive law)

h. $\phi \vee (\psi \wedge \chi) \Leftrightarrow (\phi \vee \psi) \wedge (\phi \vee \chi)$ (distributive law)

i. $\phi \rightarrow \psi \Leftrightarrow \neg\phi \vee \psi \Leftrightarrow \neg(\phi \wedge \neg\psi) \Leftrightarrow \neg\psi \rightarrow \neg\phi$

j. $\phi \rightarrow \psi \Leftrightarrow \neg\psi \rightarrow \neg\phi$ (law of contraposition)

k. $\phi \leftrightarrow \psi \Leftrightarrow (\phi \rightarrow \psi) \wedge (\psi \rightarrow \phi) \Leftrightarrow (\phi \wedge \psi) \vee (\neg\phi \wedge \neg\psi)$

l. $(\phi \vee \psi) \wedge \neg(\phi \wedge \psi) \Leftrightarrow \neg(\phi \leftrightarrow \psi) \Leftrightarrow \neg\phi \leftrightarrow \psi$

m. $(\phi \vee \psi) \rightarrow \chi \Leftrightarrow (\phi \rightarrow \chi) \wedge (\psi \rightarrow \chi)$

n. $\phi \rightarrow (\psi \wedge \chi) \Leftrightarrow (\phi \rightarrow \psi) \wedge (\phi \rightarrow \chi)$

o. $\phi \rightarrow (\psi \rightarrow \chi) \Leftrightarrow (\phi \wedge \psi) \rightarrow \chi$

Equivalences and interchangeability

Observations:

- Only the truth value of a formula matters (referentiality)
- The truth value of a complex is determined by the truth values of the components (compositionality)
- Equivalent formulas always have the same truth value

Conclusion:

- Equivalent formulas make the same contribution to the truth value of a complex

So: equivalent formulas are interchangeable

Notation

We use the following notation

- $[\phi/\psi]\chi$ is the formula χ' that results from replacing every occurrence of ψ in χ by ϕ

Examples:

- $[p/q]q = p$
- $[p/q]q \to p = p \to p$
- $[p/q](p \wedge r) = p \wedge r$ ('empty')
- $[(p \wedge q)/r]((r \wedge s) \to (r \vee s) = ((p \wedge q) \wedge s) \to ((p \wedge q) \vee s)$
- $[p/q][r/s](s \vee \neg q) = [p/q](r \vee \neg q) = r \vee \neg p$ ('stacking')

Substitution theorem

Theorem (Substitution)

If ϕ and ψ are logically equivalent, then χ is logically equivalent with $[\phi/\psi]\chi$

Proof.

By induction on the construction of χ. Observation: ϕ, ψ, χ are complex: proposition letters are by definition not equivalent and do not have sub-formulas.

- $\chi = \neg\psi$. Then $[\phi/\psi]\chi = \neg\phi$. Since for all V: $V(\phi) = V(\psi)$, we also have that for all V: $V(\neg\phi) = V(\neg\psi)$
- $\chi = \psi \wedge \xi$. Then $[\phi/\psi]\chi = \phi \wedge \xi$. Since for all V: $V(\phi) = V(\psi)$, we also have that for all V: $V(\psi \wedge \xi) = V(\phi \wedge \xi)$

[Other cases by similar reasoning] □

Using equivalences

Given the Substitution Theorem we can use equivalence to prove other equivalences

Example:
$\neg\neg\phi \to \neg\neg\psi \Leftrightarrow \neg\psi \to \neg\phi$

Proof.

- We know that $\neg\neg\phi \Leftrightarrow \phi$ (Law of Double Negation)
- Given the Substitution theorem it follows that
  $[\phi/\neg\neg\phi][\psi/\neg\neg\psi](\neg\neg\phi \to \neg\neg\psi) \Leftrightarrow$
  $[\phi/\neg\neg\phi](\neg\neg\phi \to \psi) \Leftrightarrow$
  $\phi \to \psi$
- From which it follows that
  $\neg\neg\phi \to \neg\neg\psi \Leftrightarrow \phi \to \psi \Leftrightarrow \neg\psi \to \neg\phi$

□

Of properties and relations

Tautology, contradiction, contingency

Tautology

Definition (Tautology)

A formula ϕ is a tautology iff for all V: $V(\phi) = 1$

Notation:

- '$\models \phi$' stands for 'ϕ is a tautology'

Fact

If ϕ and ψ are tautologies, then $\phi \Leftrightarrow \psi$

Notation:

- '$\top$' stands for an arbitrary tautology

$\models \phi \vee \neg\phi$ ('Law of Excluded Middle')

ϕ	$\neg\phi$	$\phi \vee \neg\phi$
1	0	1
0	1	1

Contradiction

Definition (Contradiction)

A formula ϕ is a contradiction iff for all V: $V(\phi) = 0$

Fact

If ϕ and ψ are contradictions, then $\phi \Leftrightarrow \psi$

Notation:

- '$\bot$' stands for an arbitrary contradiction

$\phi \wedge \neg\phi$

ϕ	$\neg\phi$	$\phi \wedge \neg\phi$
1	0	0
0	1	0

Contingency

Definition (Contingency)

A formula ϕ is a contingency iff for there is V: $V(\phi) = 1$ and there is V': $V'(\phi) = 0$

Of properties and relations

Connections

Tautologies and contradictions

Fact

ϕ is a tautology iff $\neg\phi$ is a contradiction

Proof.

Suppose ϕ is a tautology. Then for every V: $V(\phi) = 1$. But then for every V it must hold that $V(\neg\phi) = 0$. So according to the definition, $\neg\phi$ is a contradiction. $\square$

Contingencies

Fact

ϕ is a contingency iff $\neg\phi$ is a contingency

Proof.

$\Rightarrow$: Suppose ϕ is contingent. Then there is a V: $V(\phi) = 1$ and a V' : $V'(\phi) = 0$. For the first we have $V(\neg\phi) = 0$ and for the second $V'(\neg\phi) = 1$, which means that $\neg\phi$ is contingent

$\Leftarrow$: analogously

□

Validity

Semantic view

One way to define validity is as follows:

> *An argument is valid if and only if the truth of the premisses guarantees the truth of the conclusion*

Note:

- *Actual* truth or falsity of premisses and/or conclusion is irrelevant

Definition

Definition (Validity)

An argument scheme $\phi_1, \ldots, \phi_n / \psi$ is valid iff for all V: $V(\phi_1) = \cdots = V(\phi_n) = 1 \Rightarrow V(\psi) = 1$

Notation:

- '$\phi_1, \ldots, \phi_n \models \psi$ ' for '$\phi_1, \ldots, \phi_n / \psi$ is valid'

Note:

- '$\models \phi$': a tautology follows from zero premisses
- Since we have Weakening:

$$\phi \models \psi \Rightarrow \phi \wedge \chi \models \psi$$

  this means:
  a tautology follows from any arbitrary set of premisses

Checking validity: truth tables

Procedure:

- construct truth table for premisses and conclusion
- check conclusion in all rows in which all premisses are 1:
  - if the conclusion is 1, the argument scheme is valid
  - if in (at least) one row the conclusion is 0, the argument is invalid

A row in which all premisses are 1 and the conclusion is 0 is called a counterexample

Example: validity

$p \to (q \wedge r),\ q \to \neg r \models \neg p$

p	q	r	$q \wedge r$	$p \to (q \wedge r)$	$\neg r$	$q \to \neg r$	/	$\neg p$
1	1	1	1	1	0	0		
1	1	0	0	0	1	1		
1	0	1	0	0	0	1		
1	0	0	0	0	1	1		
0	1	1	1	1	0	0		
0	1	0	0	1	1	1	*	1
0	0	1	0	1	0	1	*	1
0	0	0	0	1	1	1	*	1

Example: counterexample

$p \to (q \wedge r),\ \neg r \to q \not\models \neg p$

p	q	r	$q \wedge r$	$p \to (q \wedge r)$	$\neg r$	$\neg r \to q$	/	$\neg p$
1	1	1	1	1	0	1	*	0
1	1	0	0	0	1	1		
1	0	1	0	0	0	1		
1	0	0	0	0	1	0		
0	1	1	1	1	0	1	*	1
0	1	0	0	1	1	1		
0	0	1	0	1	0	1		
0	0	0	0	1	1	0		

Proving validity: valuations

$p \to (q \wedge r),\ q \to \neg r \models \neg p$

Proof.

Suppose the argument is not valid. Then there is a V such that:

$$V(p \to (q \wedge r)) = V(q \wedge \neg r) = 1, V(\neg p) = 0 \text{ (assumption)}$$
$$\text{then } V(p) = 1 \ (\neg)$$
$$\text{so } V(q \wedge r) = 1 \text{ (first premise, } \to)$$
$$\text{then } V(q) = V(r) = 1 \ (\wedge)$$
$$V(q) = 1 \text{ implies that } V(\neg r) = 1 \text{ (second premise, } \to)$$
$$\text{so } V(r) = 0 \ (\neg)$$
$$\text{That contradicts } V(r) = 1$$

So the assumption must be rejected.

Note:

- This proof uses *reductio ad absurdum*

Counterexamples: valuations

Strategy:

Specify a V for all the proposition letters in premisses and/or conclusion such that:

- $V(\phi_i) = 1$ for all premisses ϕ_i
- $V(\psi) = 0$ for the conclusion ψ

Example:
$p \to (q \wedge r),\ \neg r \to q \not\models \neg p$:

Any V such that $V(p) = V(q) = V(r) = 1$ is a counterexample

For any such V we have:

- $V(p \to (q \wedge r)) = 1$ since $V(q \wedge r) = 1$ because $V(q) = V(r) = 1$
- $V(\neg r \to q) = 1$ since $V(q) = 1$
- $V(\neg p) = 0$ since $V(p) = 1$

Functional completeness

A meta-logical question

Logic versus meta-logic:

- A logic is a system that is used to represent and study arguments
- Meta-logic studies the properties of such system

A typical meta-logical questions is this:

> *What connective(s) do we need to express all proposition logical arguments?*

Consider the set of connectives we have: $\neg, \vee, \wedge, \rightarrow, \leftrightarrow$

Two questions:

1. Do we need all elements of the set?
2. Can all proposition logical arguments be expressed using this set?

Reducing the numbers

Consider the following equivalences:

1. $(\phi \leftrightarrow \psi) \Leftrightarrow ((\phi \rightarrow \psi) \wedge (\psi \rightarrow \phi))$
   this allows us to eliminate $\leftrightarrow$
2. $(\phi \rightarrow \psi) \Leftrightarrow (\neg\phi \vee \psi)$
   with this we can eliminate $\rightarrow$
3. $(\phi \vee \psi) \Leftrightarrow (\neg(\neg\phi \wedge \neg\psi))$
   which makes $\vee$ superfluous

So, we can make do with just $\{\neg, \wedge\}$

Using other equivalences we can also reduce the full set to, for example, $\{\neg, \rightarrow\}$

So the answer to the first question is: 'No'.

Truth functions

What about the second question?

Connectives stand for truth functions

A truth function takes one or more truth values as argument(s) and delivers a truth value as its value.

These are the truth functions that the connectives in the full set stand for:

$$f_{\neg}(1) = 0 \text{ and } f_{\neg}(0) = 1 \tag{1}$$

$$f_{\wedge}(1,1) = 1 \text{ and } f_{\wedge}(1,0) = f_{\wedge}(0,1) = f_{\wedge}(0,0) = 0 \tag{2}$$

$$f_{\vee}(1,1) = f_{\vee}(1,0) = f_{\vee}(0,1) = 1 \text{ and } f_{\vee}(0,0) = 0 \tag{3}$$

$$f_{\rightarrow}(1,1) = f_{\rightarrow}(0,1) = f_{\rightarrow}(0,0) = 1 \text{ and } f_{\rightarrow}(1,0) = 0 \tag{4}$$

$$f_{\leftrightarrow}(1,1) = f_{\leftrightarrow}(0,0) = 1 \text{ and } f_{\leftrightarrow}(1,0) = f_{\leftrightarrow}(0,1) = 0 \tag{5}$$

Semantics reformulated

Definition (Semantics)

A valuation V is a function from FORM_L to $\{0, 1\}$ that meets the following requirements:

1. $V(p_i) \in \{0, 1\}$, for all $p_i \in \text{VOC}_L$
2. $V(\neg\phi) = f_\neg(V(\phi))$
3. $V(\phi \wedge \psi) = f_\wedge(V(\phi), V(\psi))$
4. $V(\phi \vee \psi) = f_\vee(V(\phi), V(\psi))$
5. $V(\phi \to \psi) = f_\to(V(\phi), V(\psi))$
6. $V(\phi \leftrightarrow \psi) = f_\leftrightarrow(V(\phi), V(\psi))$

More truth functions

Note we also have 3-place truth functions, e.g.:

$$f(1,1,1) = f(0,0,0) = 1 \text{ and}$$
$$f(1,1,0) = f(1,0,0) = f(1,0,1) = f(0,1,1) = f(0,1,0) = f(0,0,1) = 0$$

And so on ...

If fact, for any n we have 2^{2^n} truth functions.

So the question:

- Can all proposition logical arguments be expressed using the standard set?

boils down to this one:

- Can all truth functions, i.e., all functions f: $\{0,1\}^n \to \{0,1\}$, for arbitrary n, be expressed by $\{f_\neg, f_\wedge, f_\vee, f_\to, f_\leftrightarrow\}$?

Functional completeness

The answer is ‘Yes’:

Theorem (Functional completeness of proposition logic)

For all n-ary truth functions f, there is a formula ϕ with n proposition variables $p_1, \ldots, p_n$ such that for every valuation V of $p_1, \ldots, p_n$, $V(\phi) = f(V(p_1), \ldots, V(p_n))$

We don’t do the full proof, but illustrate how it works by considering some examples.

1-place truth functions

There are four 1-place truth functions:

x	$f_1(x)$	$f_2(x)$	$f_3(x)$	$f_4(x)$
1	1	1	0	0
0	1	0	1	0

These are expressible by:

1. $p \vee \neg p$
2. p
3. $\neg p$
4. $p \wedge \neg p$

Another example

Consider the 3-place function we introduced earlier:

x	y	z	$f(x, y, z)$
1	1	1	1
1	1	0	0
1	0	1	0
1	0	0	0
0	1	1	0
0	1	0	0
0	0	1	0
0	0	0	1

Note $f(x, y, z) = 1$ in two cases:

1. if $x = y = x = 1$
2. and if $x = y = x = 0$

For the first case we can write $p \wedge q \wedge r$ and the second is represented by $\neg p \wedge \neg q \wedge \neg r$.

The required formula is the disjunction of these two:

$$(p \wedge q \wedge r) \vee (\neg p \wedge \neg q \wedge \neg r)$$

Procedure

This is the general procedure:

- For each row in which the result is 1 construct a formula ϕ_i, using $\neg$ and $\wedge$, that is 1 exactly when $p_1, \ldots, p_n$ have the values that are assigned to them in that row
- The required ϕ is the disjunction of these sub-formulas ϕ_i

Using this we can prove that the set $\{\neg, \wedge, \vee\}$ is functionally complete

Note, given the equivalences noted above, we can then also prove that, e.g., $\{\neg, \vee\}$, $\{\neg, \rightarrow\}$, ...are functionally complete

Homework

Homework

- Read chapter 7
- Make exercises 1(b,i,m,o), 2(d,e), 3(c,e); 4, 5(a,d,g), 6, 7, 9(f,j,q), 10, 11, 12, 13
- Read chapter 8
- Read Harman, *Change in View*, chapters 1 and 2